Vector Triangles and Their Simple Applications [7 pts]

Q1. Area in terms of range

Let the flight time be t. Resolving the initial velocity,

$$\vec{\mathbf{v}}_0 = (v_0 \cos\theta, v_0 \sin\theta), \qquad \vec{\mathbf{g}}t = (0, -gt).$$

The cross-product magnitude of any two sides of the triangle gives (twice) its area. Choosing $\vec{\mathbf{v}}_0$ and $\vec{\mathbf{g}}t$,

$$2S = |\vec{\mathbf{v}}_0 \times (\vec{\mathbf{g}}t)| = |(v_0\cos\theta)(-gt) - (v_0\sin\theta)(0)| = gtv_0\cos\theta.$$

Since $x = v_0\cos\theta \cdot t$ is precisely the horizontal range, hence

$$\boxed{S = \tfrac{1}{2}gx}.$$

Q2. Area in angular form

For ground-to-ground motion, the vertical displacement is zero:

$$0 = v_0\sin\theta \cdot t - \tfrac{1}{2}gt^2 \implies t = \frac{2v_0\sin\theta}{g}.$$

Substituting into the area formula of Q1,

$$S = \tfrac{1}{2}\, g\,(v_0\cos\theta)\,\frac{2v_0\sin\theta}{g} = \boxed{\tfrac{1}{2}\, v_0^2 \sin 2\theta}.$$

Now consider the same triangle but with the two sides $\vec{\mathbf{v}}_0$ and $\vec{\mathbf{v}}$ meeting at the origin. For ground-to-ground motion, energy conservation gives $|\vec{\mathbf{v}}| = |\vec{\mathbf{v}}_0| = v_0$. The (signed) area of the triangle is

$$S = \tfrac{1}{2}\, |\vec{\mathbf{v}}_0|\, |\vec{\mathbf{v}}|\, \sin\alpha = \tfrac{1}{2}\, v_0^2 \sin\alpha,$$

where α is the interior angle at the origin, taken from $\vec{\mathbf{v}}_0$ to $\vec{\mathbf{v}}$. At landing $\vec{\mathbf{v}} = (v_0\cos\theta,\, -v_0\sin\theta)$, so $\alpha = -2\theta$ (the velocity has rotated clockwise by 2θ). Hence

$$\sin\alpha = \sin(-2\theta) = -\sin 2\theta, \qquad S = -\tfrac{1}{2}\, v_0^2 \sin\alpha.$$

The two forms are therefore equivalent: $\boxed{S = \tfrac{1}{2}v_0^2\sin 2\theta = -\tfrac{1}{2}v_0^2\sin\alpha}$.

The maximum of S — and therefore of x — occurs when $\sin 2\theta = 1$, i.e. $\boxed{\theta = 45^\circ}$, recovering the familiar result through a purely geometric route.

Q3. Non-zero launch height

The mass is launched from $y = h > 0$ and lands at $y = 0$. The flight time satisfies

$$0 = h + v_0\sin\theta \cdot t - \tfrac{1}{2}gt^2,$$

giving

$$t(\theta) = \frac{v_0\sin\theta + \sqrt{v_0^2\sin^2\theta + 2gh}}{g}.$$

The horizontal range is $L(\theta) = v_0\cos\theta \cdot t(\theta)$.

To maximise L, set $\mathrm{d}L/\,\mathrm{d}\theta = 0$. Writing L explicitly and differentiating (or, more elegantly, using the implicit relation $L\tan\theta - gL^2/(2v_0^2\cos^2\theta) + h = 0$ obtained by eliminating t), one finds

$$\boxed{\sin^2\theta_{\mathrm{opt}} = \frac{v_0^2}{2(v_0^2 + gh)}}.$$

Equivalently,

$$\theta_{\mathrm{opt}} = \arcsin\sqrt{\frac{v_0^2}{2(v_0^2 + gh)}}, \qquad \text{or} \qquad \tan\theta_{\mathrm{opt}} = \frac{v_0}{\sqrt{v_0^2 + 2gh}}.$$

Geometric note. From Q1, $S = \frac{1}{2}gL$ still holds. Energy conservation gives $|\vec{\mathbf{v}}|^2 = v_0^2 + 2gh$, so the two sides meeting at the origin have *fixed* lengths v_0 and $\sqrt{v_0^2 + 2gh}$. The area $S = \frac{1}{2}v_0\,|\vec{\mathbf{v}}|\sin\varphi$ (where φ is the angle between $\vec{\mathbf{v}}_0$ and $\vec{\mathbf{v}}$) is maximised when $\varphi = 90^\circ$, i.e. when $\vec{\mathbf{v}}_0 \perp \vec{\mathbf{v}}$. Imposing orthogonality on the vector triangle yields the same optimal angle as above, without calculus.